Technical Report

U.S. Criteria Air Pollutants, 2021–2025

An EPA AQS Statistical Analysis in R

Prepared by

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Data: U.S. EPA Air Quality System (AQS), AirData files dated 25 June 2026

Code: github.com/Yousef-sham5/houston-dfw-air-quality

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Reproducibility

Data: U.S. EPA Air Quality System, file versions dated 25 June 2026. Analysis done in R 4.6.1 and R Markdown, with package versions saved in renv.lock. The numbers in the text are pulled straight from output tables written by the analysis scripts, and all figures are made in code, so the full report can be rebuilt from the raw data. Code: github.com/Yousef-sham5/houston-dfw-air-quality.

01 Introduction

The U.S. Environmental Protection Agency (EPA) sets National Ambient Air Quality Standards (NAAQS) for six criteria pollutants, with limits chosen to protect public health [1]. Many areas still do not meet them. In 2023, 42.7% of ozone monitoring sites and 39.0% of annual PM2.5 sites with data in all five study years were above their standards (Section 3).

Houston and Dallas–Fort Worth both have long-running ozone problems. EPA classifies both metros as severe nonattainment areas for the 2008 ozone standard and serious nonattainment areas for the 2015 standard [2, 3]. In this dataset they rank 6th and 8th of 353 metros with an ozone exceedance, by exceedance site-days. This count goes up with the number of monitors in a metro, so it is only a rough comparison. Houston also has the Houston Ship Channel. The American Society of Civil Engineers calls the area around it the second largest petrochemical complex in the world [4], with refineries, chemical plants and ships along its banks.

This study compares air near the Ship Channel with the rest of Harris County ("Houston other") and with Dallas–Fort Worth, and checks whether the differences match basic atmospheric chemistry. It tests three hypotheses:

> **H1.** NO_2 is higher near the channel. Process heaters, flares, ship engines and diesel trucks all burn fuel and give off nitrogen oxides, so sites close to many of these sources should read higher.
>
> **H2.** Ozone is lower near the channel because fresh NO reacts with it in the titration reaction $NO + O_3 \rightarrow NO_2 + O_2$. Each ozone molecule used up makes one NO_2 molecule, so this reaction does not change the sum $Ox = O_3 + NO_2$. If titration explains the ozone gap, Ox should be about the same in both groups.
>
> **H3.** SO_2 comes from short, local plumes. If so, it should show up as brief spikes when the wind blows from certain directions, with the other hours near the detection limit.

The analysis uses EPA Air Quality System (AQS) data from 2021 to 2025 at two scales. Daily PM2.5, ozone, NO_2, SO_2 and CO data from across the U.S. give the national picture. The case study pairs hourly ozone, NO_2 and SO_2 from 16 monitors in Houston and Dallas–Fort Worth with hourly wind and temperature. The Ship Channel group is the 9 Harris County sites within 7.5 km of the channel's land-cut sections. Each hourly group has only two ozone and NO_2 sites, so the hourly results describe those sites only. Each comparison is rerun with other site groups and data subsets to see if it still holds. Observational data cannot prove what caused a pattern, so links to a source are written as "consistent with" that source.

Section 2 covers the data and methods. Section 3 gives the national context, and Sections 4 and 5 cover the daily and hourly case study. Section 6 looks at the shape of the daily values, fits curves to them, checks how well the curves count high days, and works through one ozone design value using EPA's rules. Section 7 lists the limitations, and Section 8 returns to H1 to H3.

02 Data and methods

02.1 Data and cleaning

The data are EPA's AirData pre-generated files [5], in the versions dated 25 June 2026: 25 daily, 15 hourly pollutant and 10 hourly weather files. Table 2.1 lists the parameter codes, daily values and standards. Only PM2.5 from federal reference and equivalent methods (FRM and FEM) is used. The weather data are AQS wind speed, wind direction and temperature (61103, 61104, 62101).

Table 2.1. Pollutants, AQS parameter codes, daily values and the NAAQS each site-year was screened against [1].

Pollutant	AQS code	Daily value	Standard screened
PM2.5	88101	24-hour mean	35 µg/m^3, 98th percentile day
PM2.5 (annual)	88101	Annual mean	9.0 µg/m^3, mean of four quarters
Ozone	44201	Maximum 8-hour mean	0.070 ppm, 4th-highest day
NO_2	42602	Maximum 1-hour value	100 ppb, 98th percentile day
SO_2	42401	Maximum 1-hour value	75 ppb, 99th percentile day
CO	42101	Maximum 8-hour mean	9 ppm, 2nd-highest day

Cleaning cut the daily files from 8,930,992 to 5,557,574 rows. Most removed rows were duplicate summaries of the same monitor and day (Table A.1, Appendix). Missing data are never filled in, and days with fewer than 75% of hours reported are dropped. Other unusual data are flagged and kept, so each result can be rerun without them. These include event days, hourly readings below the detection limit or on incomplete days, one monitor move and several method changes. Event days come from a table of 203 state and county records, each with a source. The 79,402,201 raw hourly rows were read in chunks and cut to 661,545 monitor-hours for the 16 case-study monitors. These hours rebuild all 27,747 daily maximums in EPA's daily files exactly. Times are in local standard time, as AQS reports them.

02.2 Site groups

The Ship Channel group is the Harris County sites within 7.5 km of the channel's 12 land-cut sections in the U.S. Army Corps of Engineers channel framework [6]. The cutoff falls between Houston Deer Park #2 (6.94 km) and Houston North Wayside (8.49 km). The rest of Harris County is "Houston other". Seabrook is left out because it sits next to the separate Bayport industrial district. Dallas and Tarrant counties form Dallas–Fort Worth. Table 2.2 gives the sites in each group, and Section 4 maps the hourly sites.

Table 2.2. Sites per group. Daily counts are sites with data in all five years; hourly counts are the case-study monitors, which do not include PM2.5.

	Daily, five-year sites			Hourly monitors		
Pollutant	**Ship Channel**	**Houston other**	**DFW**	**Ship Channel**	**Houston other**	**DFW**
PM2.5	3	1	3	–	–	–
Ozone	6	10	6	2	2	2
NO_2	4	3	4	2	2	2
SO_2	3	1	0	3	1	0

02.3 How comparisons are tested

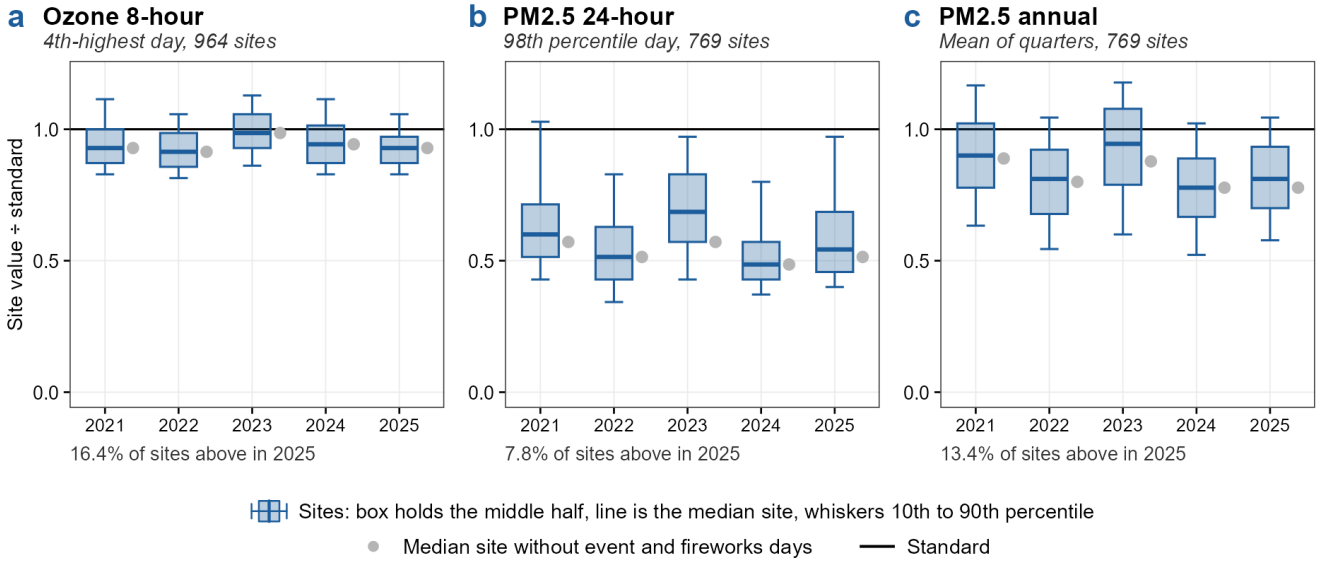
Groups are compared with same-day differences. Each day, a group's value is the mean of its sites that reported, and the two groups are subtracted on the same day. Both groups share that day's weather, so most of the weather effect cancels out. The median daily difference gets a 95% range from a block bootstrap that resamples 14-day blocks 1,000 times. Blocks are used because neighbouring days are correlated. Each result is then rerun under up to eight checks. These include other site groups, such as the one in TCEQ's 2024 air toxics memo [7], adding four-year sites, and dropping event days or days after a monitor change. A check that does not change the data is skipped. A result holds when it keeps its sign and its range does not include zero. Appendix A.3 shows the code.

03 National context

03.1 Network and screening

Level comparisons use only sites with data in all five years (964 ozone, 769 PM2.5, 312 NO_2, 301 SO_2 and 135 CO sites). Exceedance counts use all sites. The number of reporting SO_2 and CO sites fell by 18.1% and 17.4%, and PM2.5 sites grew by 3.9%. The share of PM2.5 data from FEM monitors rose from 79.2% to 85.2%. At the same site and day, FEM read 0.3 to 0.5 µg/m^3 higher than FRM (Figure 2A.1, Appendix).

Each site-year was checked against its standard in the form and rounding the regulations give [8–12] (Figure 2A.2). Ozone and annual PM2.5 fail most often. In 2023, 42.7% of ozone sites and 39.0% of annual PM2.5 sites were above their levels. The 9.0 µg/m^3 annual PM2.5 level, set in 2024, is applied to every year. With three-year windows, which are closer to how EPA judges compliance, 22.8 to 24.1% of ozone sites were above the standard.



Source: EPA AQS daily data, 2021 to 2025, sites with data in all five years. Single-year screening values, not EPA design values.

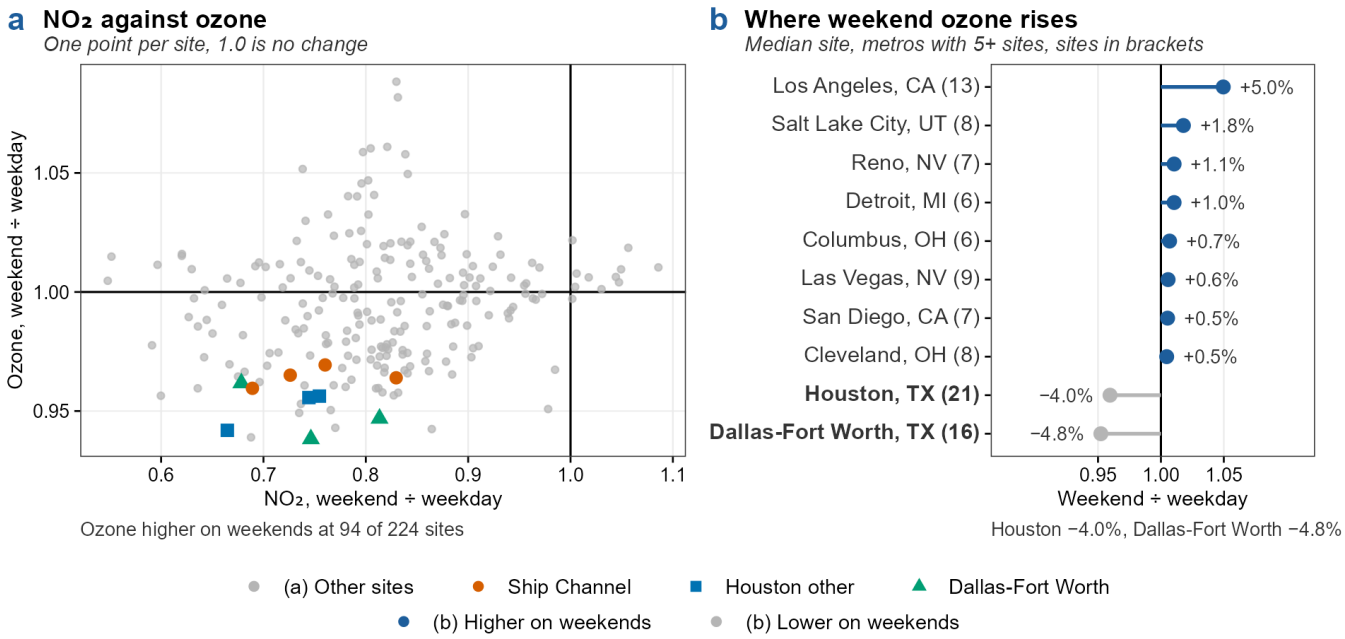
Figure 2A.2. Ozone and annual PM2.5 fail their standards most often.
Record: national_naaqs_screening.csv

03.2 Events and metro ranks

Events such as smoke, dust and storms matter much more for PM2.5 than for ozone (Figure 2A.4, Appendix). Event days hold 39.9% of PM2.5 exceedance site-days, and 68.2% in 2023, the year of the large Canadian wildfire smoke. Only 14.4% of these event-day exceedances carry EPA's exceptional-event flag. For ozone the share is 1.2%, consistent with ozone exceedances coming mostly from normal summer photochemistry. By exceedance site-days, Houston and Dallas–Fort Worth rank 6th and 8th of 353 metros for ozone, and 52nd and 89th of 359 for PM2.5 (Figure 2A.5, Appendix). Ozone is the bigger problem in both metros.

03.3 Seasons and the weekend effect

Ozone peaks in June at 1.58 times its lowest month, and PM2.5 peaks in July at 1.51 times. This is consistent with summer photochemistry and, for PM2.5, wildfire smoke. NO_2 and CO peak in winter at 1.75 and 1.73 times their lows, consistent with a shallow winter mixing layer that holds fresh emissions near the ground. From April to October, weekend NO_2 at the median site drops to 0.805 of its weekday level, while ozone stays at 0.987. Ozone does not track NOx one for one. Where VOCs limit ozone formation, less NO on weekends means less titration, so ozone can go up. Weekend ozone is 5% higher than weekday ozone in Los Angeles, consistent with VOC-limited conditions. It is 4% and 4.8% lower in Houston and Dallas–Fort Worth, consistent with more NOx-limited summer weekends (Figure 2A.10). Section 5 looks at titration again with hourly data.



Source: EPA AQS daily data, April to October 2021 to 2025, sites with data in all five years, without event and fireworks days. Weekend: Saturday and Sunday; Memorial Day, Independence Day and Labor Day are left out.

Figure 2A.10. Weekend ozone rises in Los Angeles and falls in Houston and Dallas–Fort Worth.
Record: national_weekend_summary.csv, national_ozone_weekend_metros.csv

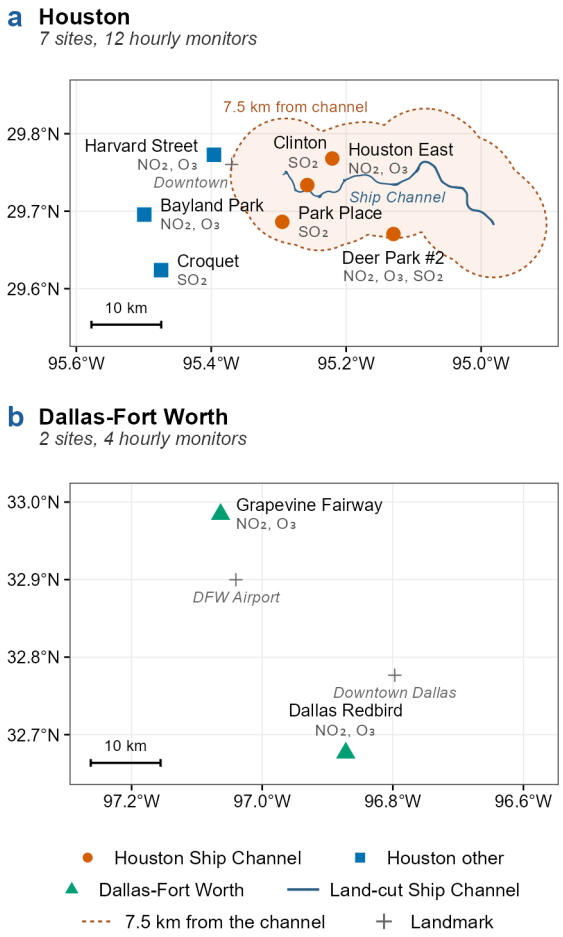
04 Houston case study: daily

This section compares the three site groups using daily values at the sites with data in all five years (Table 2.2). Figure 1C.1 shows the hourly monitors used in Section 5. The daily groups include more Harris County sites, split by the same 7.5 km rule.

04.1 NO_2 and ozone by group

NO_2 is highest at the Ship Channel in all 5 years. The median Ship Channel site's mean daily NO_2 peak is 1.44 ppb above Houston other. Leaving out the days after the NO_2 method changes at Park Place (November 2023) and Harvard Street (April 2024) raises this to 2.49 ppb, mostly because Park Place, a Ship Channel site, reads lower on its new method. NO_2 also drops with distance from the channel. With four-year sites added, the Spearman correlation across 10 sites is $\rho = -0.72$ ($p = 0.02$). Dropping one site at a time keeps ρ between -0.83 and -0.62. Without the farthest site (48.9 km away), p rises to 0.077, so the distance trend is only supporting evidence.

Ozone shows the opposite pattern. Dallas–Fort Worth has the most ozone exceedance days per site in all 5 years, and the Ship Channel has the fewest in 3 of 5, consistent with fresh NO lowering ozone near the channel (H2).



Source: EPA AQS site coordinates; USACE channel. Shaded: within 7.5 km of the channel.

Figure 1C.1. Nine case-study sites in three groups.
Record: data/reference/case_study_sites.csv

04.2 Same-day differences

Table 4.1 gives the same-day differences from Section 2.3. On the same days, the Ship Channel has 1.15 ppb more NO_2 and 0.97 ppb less ozone than Houston other. This is the pattern titration predicts, and Section 5 tests it directly with Ox. Houston as a whole is 4.00 ppb lower in ozone and 1.77 ppb higher in NO_2 than Dallas–Fort Worth, and both results hold in every run.

PM2.5 and SO_2 show no useful group gap. The PM2.5 difference holds in only 3 of 6 runs, consistent with PM2.5 being mostly regional. The SO_2 difference holds in every run, but at 0.13 ppb it is far below the 2 ppb detection limit. Section 5 uses SO_2 spike hours instead. Croquet, in Houston other, has more SO_2 spike days than the average Ship Channel site in all 5 years. Figure 2B.7 in the Appendix shows how much single days swing around these medians, and how the gaps change by season.

Table 4.1. Median same-day differences between groups, with block-bootstrap 95% ranges and the share of check runs in which each result holds its sign with a range clear of zero.

Comparison	Pollutant	Median difference (95% range)	Runs holding
Ship Channel minus Houston other	NO_2	+1.15 (+0.81 to +1.41) ppb	7 of 7
	Ozone	−0.97 (−1.11 to −0.77) ppb	6 of 7
	PM2.5	−0.20 (−0.40 to −0.03) µg/m^3	3 of 6
	SO_2	+0.13 (+0.10 to +0.20) ppb	4 of 4
Houston minus Dallas–Fort Worth	NO_2	+1.77 (+1.26 to +2.26) ppb	7 of 7
	Ozone	−4.00 (−4.87 to −3.27) ppb	6 of 6
	PM2.5	+0.47 (+0.22 to +0.73) µg/m^3	6 of 6

04.3 Events

Most Ship Channel monitors stopped reporting during Hurricane Beryl in July 2024. On average only 19.6% of them reported during the event window, compared with 58.1% in Houston other and 100% in Dallas–Fort Worth (Figure 2B.6, Appendix). Winter Storm Uri raised PM2.5 to 1.20 to 1.57 times its normal level across the groups.

05 Houston case study: hourly

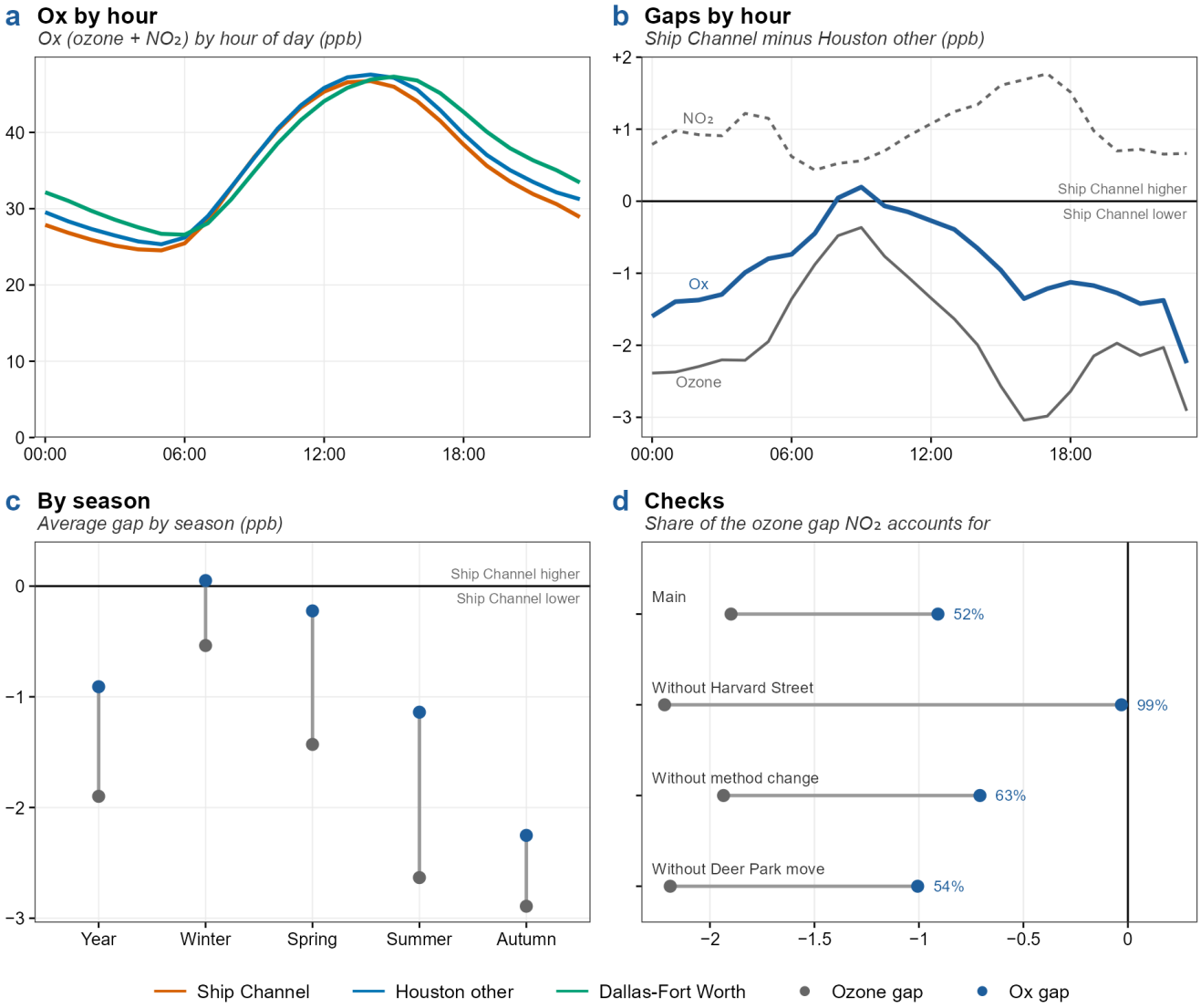
Hourly data show what time of day the group gaps appear and how they change with wind. Each group has two hourly ozone and NO_2 sites, so these results describe those sites. Harvard Street is the only Houston other site between 7.5 and 15 km from the channel. It changes the ozone result the most, so the main numbers are also given without it. All three groups report in the same hour in at least 95% of hours.

05.1 Daily cycles

At all 24 hours and under every check, Ship Channel NO_2 is higher and ozone is lower than in Houston other. Both gaps are largest late in the day, with NO_2 +1.9 ppb at 17:00 and ozone −2.96 ppb at 16:00 (Figure 2C.2, Appendix). From April to October, weekend NO_2 from 6 to 8 am is 36% below weekdays at the Ship Channel, compared with 42% in Houston other and 50% in Dallas–Fort Worth. The smaller weekend drop is consistent with more Ship Channel NO_2 coming from sources that run every day, such as industry.

05.2 The Ox balance

Titration ($NO + O_3 \rightarrow NO_2 + O_2$) turns each ozone molecule it uses up into one NO_2 molecule, so $Ox = O_3 + NO_2$ stays the same. If titration were the only reason Ship Channel ozone is lower, both groups would have the same Ox. Over the year, Ship Channel ozone is 1.90 ppb lower than Houston other (95% range 1.60 to 2.25), while its Ox is only 0.91 ppb lower. This means **Ox closes 52% of the ozone gap** (Figure 2C.4, Table 5.1). Without Harvard Street it closes 99%, and it closes at least 52% under every check. In winter, when little new ozone forms, the range of the main Ox gap includes zero. This is consistent with fresh NO from channel sources using up ozone close to where it is released. There are no NO data for these sites, so the leftover Ox gap cannot be split into local and regional parts.



Source: EPA AQS hourly data, 2021 to 2025, ordinary hours with both pollutants at the same site, 2 sites per group, each group the mean of its sites. (b) to (d) Same-hour gaps. Standard-method NO_2 also counts some other nitrogen oxides.

Figure 2C.4. Adding NO_2 back closes at least half the ozone gap.
Record: case_study_ox_checks.csv

Table 5.1. Ship Channel minus Houston other, ppb. Gap closed is the share of the ozone gap that Ox closes.

Hours	Ozone	NO_2	Ox	Gap closed
All hours, all year	−1.90	+0.99	−0.91 (−1.24 to −0.58)	52%
All hours, without Harvard Street	−2.22	+2.19	−0.03	99%
All hours, winter	−0.54	+0.59	+0.05 (−0.46 to +0.56)	109%
Wind from the channel, without Harvard Street	−3.56	+1.41	−2.16 (−2.73 to −1.66)	40%
Other wind, without Harvard Street	−2.35	+2.38	+0.03 (−0.30 to +0.35)	101%

05.3 Calm air and wind

Calm hours (wind below 0.5 m/s) make up 6.0% of hours at the Ship Channel, 8.9% in Houston other and 1.6% in Dallas–Fort Worth. Compared with windy hours at the same site, hour and month, calm hours have 3.0 to 6.8 ppb more NO_2 and 3.8 to 9.0 ppb less ozone on average at every site, and no site's 95% range includes zero. Single days vary much more (Figure 2C.8, Appendix). In every group, at least 70% of the ozone lost in calm air shows up as NO_2. This is consistent with titration in still air near the ground.

The Ox gap left over after titration shows up mostly when the wind comes from the channel. Without Harvard Street, Ship Channel Ox is 2.16 ppb lower in wind from the channel and +0.03 ppb in other wind, where the range includes zero (Table 5.1, Figure 2C.11 in the Appendix). This is consistent with ozone forming in channel air as it moves across Houston, so sites downwind of the channel gain Ox. At all four Houston sites, NO_2 is also 1.4 to 1.6 times its usual level when the wind comes from 60 to 90 degrees. Wind direction alone does not prove a source, since each direction also brings different weather.

05.4 SO_2 plumes

At three of the four sites, SO_2 is below the 2 ppb detection limit in 97.2% to 99.4% of hours, so the analysis uses spike hours of 5 ppb or more. Spikes are short, with a median length of 1 hour at every site. Croquet, in Houston other, has the most spikes, at 5.4 per 1,000 hours. Of its spikes, 96% fall between 9 am and 6 pm, and its summer spike rate is 4.1 times that of any other season. This is consistent with an elevated plume that daytime heating mixes down to the ground. Most of its spike hours (75%) come from the southwest, away from the channel, even though only 11% of its hours have wind from that direction (Figure 2C.10, Appendix). At Park Place, 77% of spike hours come from the channel side, which accounts for 30% of its hours. During spike hours at Deer Park #2, NO_2 is 7.5 ppb above normal (5.1 to 10.0), so the plumes also carry nitrogen oxides. These patterns support H3: SO_2 comes in short, local plumes from more than one direction.

05.5 Robustness

All 10 main hourly findings hold under every check that applies. The checks drop Harvard Street, its hours after the method change, Deer Park #2 after its move, and hours on the new wind sensors (Figure 2C.12, Appendix). All 67 bootstrap ranges were also rerun with a second random seed, and none changed whether its range crosses zero.

06 Distributions

Sections 3 to 5 compare average levels. This section looks at three more things that matter before the data go into a model: the shape of the daily values, whether the groups differ in how much ozone changes from day to day, and whether a fitted curve counts high days correctly. At each five-year site, ordinary days are divided by the site's mean and fitted with three common skewed distributions (lognormal, gamma and Weibull), plus a normal distribution for comparison. The three skewed distributions each have two parameters, so AIC, a standard measure of fit, can compare them fairly. Sites where a quarter or more of days are at or below zero are left out. Almost all of these are SO_2 sites, and 2,425 sites remain. A curve only counts as the clear best fit if it still wins when the data are resampled in two-week blocks.

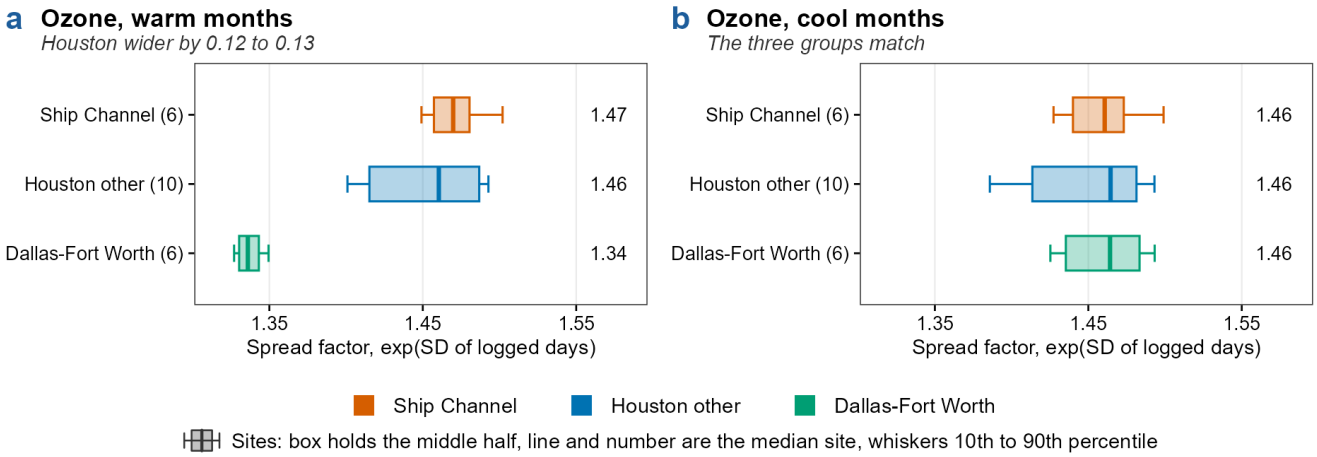
06.1 Shape and fitted curves

Each site is fitted on its own. Pooling all sites together hides the shape, because much of the spread would then come from differences between sites instead of day-to-day changes. At almost every PM2.5 and SO_2 site, and at most CO sites, the values are right-skewed. A few high days sit far above the rest, and a lognormal curve fits better than a normal one. Ozone is close to symmetric, and a normal curve fits better than a lognormal at 82% of ozone sites. NO_2 is in between, with no single shape clearly best (Figure 3.5, Appendix).

These best-fit labels are less certain than they look. Neighbouring days are correlated, so five years of daily data carry less independent information than the number of days suggests. After accounting for this, the best curve for ozone stays clearly best at only about half of the sites tested. For PM2.5, the monitor type also matters. The same sites look much more lognormal on a T640 monitor than on other monitors, consistent with an instrument effect, but the cause is not confirmed. This means log-scale summaries suit PM2.5, SO_2 and CO, ozone can be summarised on its normal scale, and PM2.5 comparisons across years should note when a site changed monitors.

06.2 Ozone spread in the case study

Figure 3.6 compares how much ozone changes from day to day at each site. It uses a spread factor, which is how many times above or below its usual level a site's ozone typically is. A larger factor means bigger day-to-day changes. In warm months, the spread factor at Houston sites is 0.12 to 0.13 higher than at Dallas–Fort Worth sites, and neither range includes zero. The Ship Channel and Houston other are about the same, and in cool months no group differs. So near the channel, ozone is lower on average (Sections 4 and 5) while its day-to-day spread stays the same as in the rest of Houston. The warm-month gap covers all of Houston. This is consistent with Houston switching between clean Gulf air and still, sunny days, but that cause is not tested here.



Source: EPA AQS daily data, 2021 to 2025, ordinary days, case-study sites, site counts in brackets; cool-season ozone on year-round sites. Warm-month gaps to Dallas-Fort Worth have 95% ranges clear of zero from 1,000 two-week block resamples.

Figure 3.6. Warm-month ozone varies more at Houston sites.
Record: distribution_fit_checks.csv

06.3 Counting high days

A curve can fit typical days well and still get the rare high days wrong. This matters for any model that estimates how often a limit is exceeded. At most sites, the best-fit curve is off by more than 25% on the number of very high days (days above the site's 99th percentile). Some of these errors cancel out when sites are added together, but for NO_2 the curves still overcount high days by 54% in total. The choice of curve also matters. At the ozone standard, the best curves match the measured count, while a lognormal curve predicts about twice as many days above the limit as were measured.

For warm-month ozone in the case study, a gamma curve gets the number of days above the standard right in all three groups, within the uncertainty of the measured count. It still overcounts the highest days (the top 1% at each site) by 51% to 69%. Readings below the detection limit also affect the fit. At Deer Park #2, treating them as "somewhere below the limit" instead of as exact values brings the predicted number of high SO_2 days from about double the measured number down to about right. The fitted curves are useful for describing the data, but high days should still be counted from the measurements.

06.4 A worked design value

An ozone design value is the three-year average of each year's 4th-highest daily maximum 8-hour value, truncated to three decimal places. A site meets the standard if its design value is 0.070 ppm or lower [11]. At Houston Aldine, the 4th-highest days were 0.078, 0.097 and 0.073 ppm in 2023, 2024 and 2025. Their average, 0.0827, truncates to 0.082 ppm, which matches EPA's published value [13]. The site also just passes EPA's completeness rule, which needs at least 90% of days to be valid on average.

Doing the same for all 22 case-study ozone sites over three periods matches EPA in 61 of 66 site-periods. Each miss is 0.001 ppm low, because the daily cleaning drops incomplete days that EPA keeps when they exceed the limit. None of the misses changes whether a site passes. All 22 sites are above the standard for 2023 to 2025, and the highest value is at Deer Park #2, a Ship Channel site. The typical design values of the three groups are nearly the same, so the Ship Channel's lower average ozone does not show up on its highest days. All 12 findings in this section hold under every check that applies (Figure 3.10, Appendix).

07 Limitations

- **No trends.** Five years is too short to separate a real trend from year-to-year changes in weather, so differences between years are described but not tested.
- **Small groups.** Each group has only two hourly ozone and NO_2 sites, and Houston other has one five-year PM2.5 site. The hourly results describe those sites and may not represent whole areas.
- **Distance and wind are indirect.** Distance to the channel's water is used in place of distance to the plants along it, and each wind direction brings its own weather. Neither one proves a source, which is why source claims say "consistent with".
- **No NO or NO_x data.** Ox shows how much of the ozone gap titration explains, but the rest of the gap cannot be split into local and regional parts.

- **Near the detection limit.** SO_2 is below 2 ppb in most hours at three of the four sites. The censored fits use method-wide detection limits, which can differ from a single monitor's own limit.
- **Cleaning choices.** Dropping days with under 75% of hours puts 5 of 66 design values 0.001 ppm below EPA's, and Section 3's single-year screening is not a design value. Wind data are missing for up to 474, 314 and 161 days in a row at Bayland Park, Grapevine Fairway and Deer Park #2.
- **Monitor changes.** PM2.5 moved toward FEM monitors, and Park Place and Harvard Street changed their NO_2 methods in November 2023 and April 2024. The checks drop the affected days, but comparisons between years still partly reflect which monitors were running.
- **Many comparisons.** Dozens of ranges are reported with no multiple-comparison correction, so the results that hold under every check are the safest to cite.
- **Curves only describe the data.** The tail checks only test the fit on measured days, so the curves are not forecasts.

08 Conclusion and regulatory implications

H1 is supported. Ship Channel NO_2 is higher than Houston other in all 5 years, on the same days (+1.15 ppb, range +0.81 to +1.41) and at all 24 hours, under every check. The drop in NO_2 with distance from the channel agrees, but it depends partly on one far site.

H2 is supported in part. Ship Channel ozone is lower on the same days (−0.97 ppb, range −1.11 to −0.77) and at every hour. Adding NO_2 back closes 52% of the hourly ozone gap, or 99% without Harvard Street, so titration explains at least half of it. The Ox gap that is left shows up in wind from the channel, consistent with ozone forming in channel air as it moves downwind.

H3 is supported. SO_2 comes as short spikes, with a median length of 1 hour, on top of a background below the detection limit. The spikes come from the southwest at Croquet (75% of its spike hours) and from the channel side at Park Place (77%), consistent with more than one local source.

The results point to three things for air quality planning, keeping in mind that this is an observational study.

Lower ozone near the channel does not mean cleaner air. Much of the ozone missing near the channel shows up as NO_2, and the leftover Ox is consistent with ozone forming downwind. Every case-study ozone site, including the Ship Channel sites, is above the 0.070 ppm standard for 2023 to 2025, and the highest design value is at Deer Park #2. This suggests that NO_x near the channel and ozone across Houston should be managed as one problem.

SO_2 should be measured hourly at more than one site. One-hour spikes from different directions get averaged out in daily means and can be missed by a single monitor. This matters because the SO_2 standard is based on 1-hour values.

Method choices change the results. Monitor changes, site groupings and the choice of curve can each shift the results enough to matter. For example, a default lognormal curve roughly doubles ozone exceedance counts, and the share of the ozone gap that Ox closes depends on whether Harvard Street is included. Analyses used for decisions should state these choices and rerun the main result under each one, as this report does.

References

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2. Texas Commission on Environmental Quality. Houston-Galveston-Brazoria ozone nonattainment status. https://www.tceq.texas.gov/airquality/sip/hgb/hgb-status, accessed 29 October 2026.
3. Texas Commission on Environmental Quality. Dallas-Fort Worth ozone nonattainment status. https://www.tceq.texas.gov/airquality/sip/dfw/dfw-status, accessed 29 October 2026.
4. American Society of Civil Engineers. Houston Ship Channel, Historic Civil Engineering Landmark. https://www.asce.org/about-civil-engineering/history-and-heritage/historic-landmarks/houston-ship-channel, accessed 29 October 2026.
5. U.S. Environmental Protection Agency. AirData pre-generated data files. https://aqs.epa.gov/aqsweb/airdata/download_files.html, file versions dated 25 June 2026.
6. U.S. Army Corps of Engineers. National Channel Framework, ChannelArea layer. https://services7.arcgis.com/n1YM8pTrFmm7L4hs/ArcGIS/rest/services/National_Channel_Framework/FeatureServer, downloaded 22 October 2026.
7. Texas Commission on Environmental Quality. Region 12 2024 air monitoring evaluation memo. https://www.tceq.texas.gov/downloads/toxicology/monitoring/air-evaluations/2024/reg12.pdf
8. 40 CFR Part 50, Appendix N (PM2.5). https://www.law.cornell.edu/cfr/text/40/appendix-N_to_part_50
9. 40 CFR Part 50, Appendix S (NO_2). https://www.law.cornell.edu/cfr/text/40/appendix-S_to_part_50
10. 40 CFR Part 50, Appendix T (SO_2). https://www.law.cornell.edu/cfr/text/40/appendix-T_to_part_50
11. 40 CFR Part 50, Appendix U (ozone). https://www.law.cornell.edu/cfr/text/40/appendix-U_to_part_50
12. 40 CFR 50.8 (CO). https://www.law.cornell.edu/cfr/text/40/50.8
13. U.S. Environmental Protection Agency. Air Quality Design Values, Ozone Design Values 2023–2025. https://www.epa.gov/air-trends/air-quality-design-values, file posted 8 June 2026.

Regulation text was read on Cornell's Legal Information Institute; the official version is the eCFR (https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-50).

Appendix

A.1 Data pipeline

Run R/0_setup.R once, then the scripts in the order of Table A.2. Each script reads only the saved outputs of earlier scripts, stops on any failed check, and documents every saved table in data_dictionary.csv. Package versions are pinned in renv.lock. A clean rerun of every script on 13 November 2026 reproduced all 137 outputs byte for byte, and the scripts from 2a on were rerun in full after the figure redesign on 14 November 2026. Full source code is in the GitHub repository.

Table A.1. Daily cleaning steps, starting from 8,930,992 raw rows.

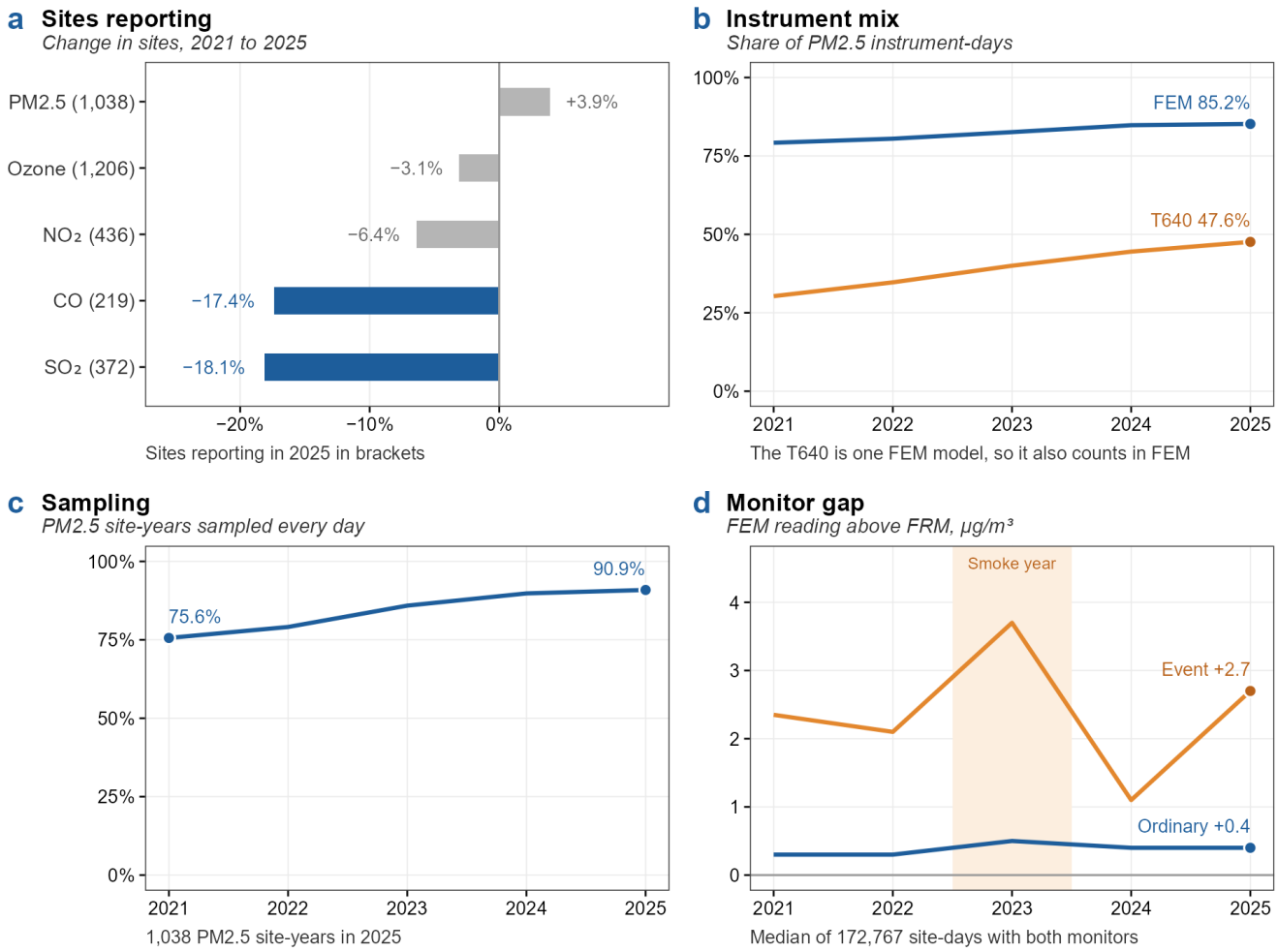
Step	Rows removed	Rows left
Drop non-US sites	9,982	8,921,010
Keep one standard per pollutant	2,956,079	5,964,931
Validate method codes	315,860	5,649,071
Keep event-included values	2,680	5,646,391
Drop T640 corrected rows overlapping firmware data	338	5,646,053
Apply 75% completeness rule	88,478	5,557,575
Drop values outside method range	1	5,557,574

Table A.2. Scripts in run order, with links to the source on GitHub.

Script	What it does	Main outputs	Lines
0_setup.R	Installs the packages and TinyTeX	None	28
utils.R	Shared helpers: chunked reader, NAAQS statistics, saving, chart theme	None	919
1a_import_clean.R	Imports and cleans the 25 daily files	Cleaned daily table, sites, coverage	1,009
1b_event_checks.R	Checks each event label against the data	None	390
1c_site_selection.R	Sets the site groups and case-study monitors	Site list, Figure 1C.1	656
1d_hourly_import_clean.R	Reads the hourly pollutant files in chunks, flags and validates them	Hourly case-study table	1,162
1e_weather_import_clean.R	Reads hourly wind and temperature, sets calm hours and wind sectors	Hourly weather table	1,450
2a_daily_eda.R	National analysis (Section 3)	Figures 2A.1 to 2A.10, national records	2,673
2b_daily_case_study.R	Daily case study (Section 4)	Figures 2B.1 to 2B.9, daily records	2,831
2c_hourly_eda.R	Hourly case study (Section 5)	Figures 2C.1 to 2C.13, hourly records	4,341
3_distributions.R	Distribution shapes, curve fits, tails and design values (Section 6)	Figures 3.1 to 3.10, distribution records	5,218

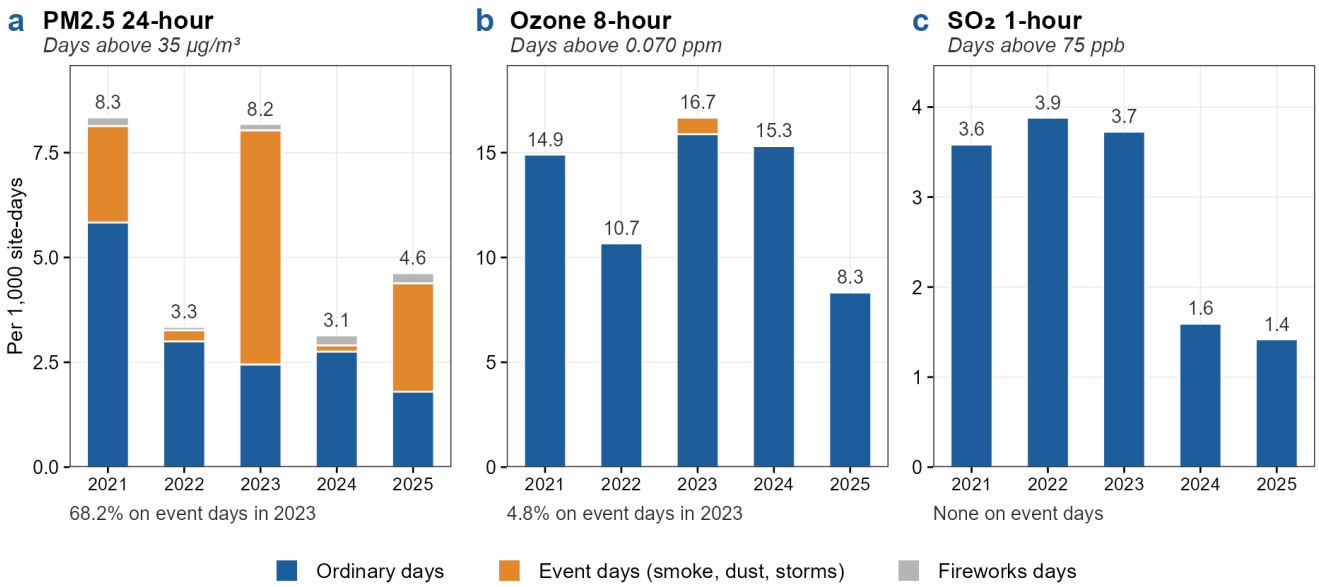
A.2 Additional figures

The 12 figures below support Sections 3 to 6, and each prints the record it is drawn from.



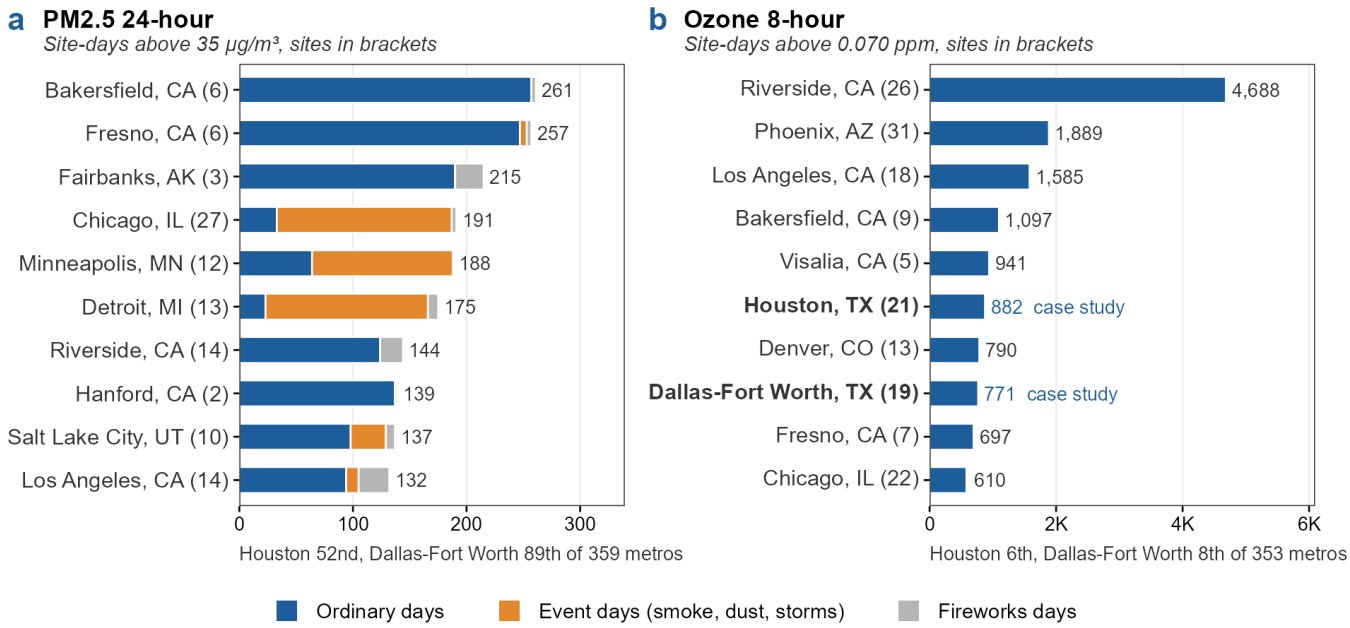
Source: EPA AQS daily data, 2021 to 2025, all sites, after Phase 1 cleaning. Event days are the labelled smoke, dust and storm windows in events.csv; fireworks days are not shown.

Figure 2A.1. SO_2 and CO sites fell as PM2.5 moved to continuous monitors.
Record: national_sites_by_year.csv, national_pm25_methods.csv



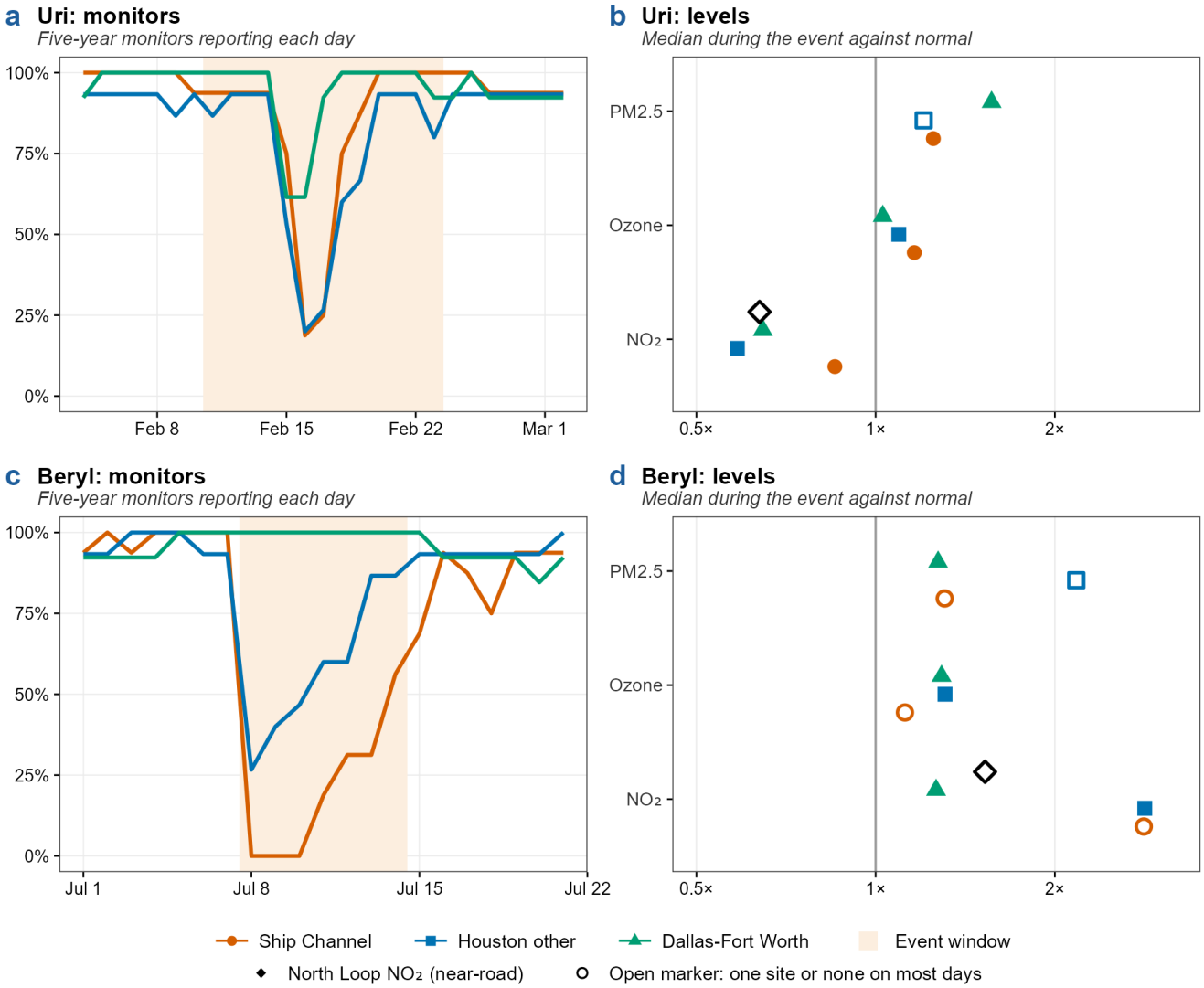
Source: EPA AQS daily data, 2021 to 2025, every site reporting that year. Screening counts, not violations: the standards allow some days above the level. Event days: events.csv.

Figure 2A.4. Event days hold over a third of PM2.5 exceedances.
Record: national_exceedances_by_year.csv



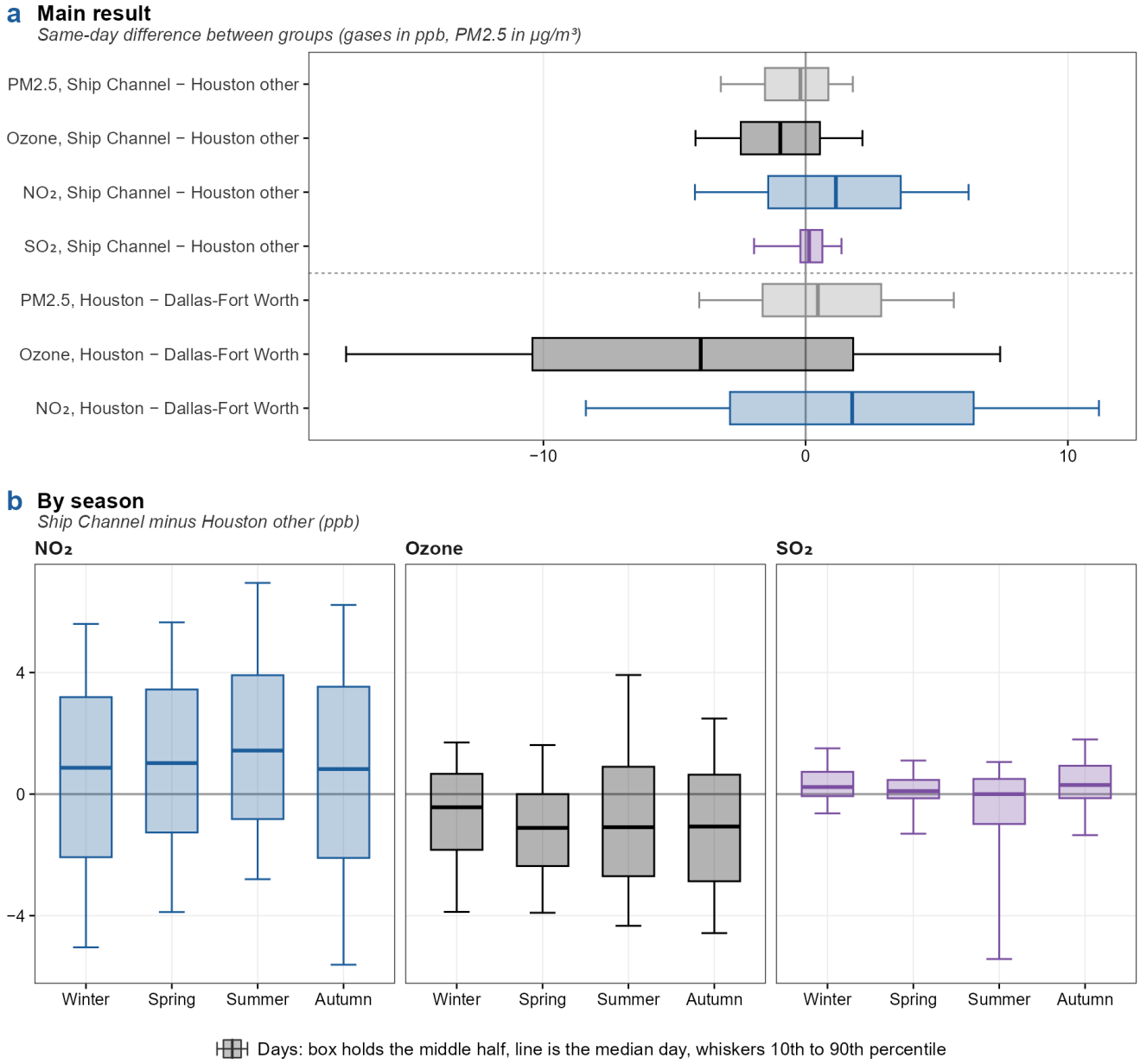
Source: EPA AQS daily data, 2021 to 2025, all sites, by core-based statistical area. Totals grow with the number of sites in a metro. Screening counts, not violations. Bold: the two case-study metros.

Figure 2A.5. California metros top both lists.
Record: national_metro_ranks.csv



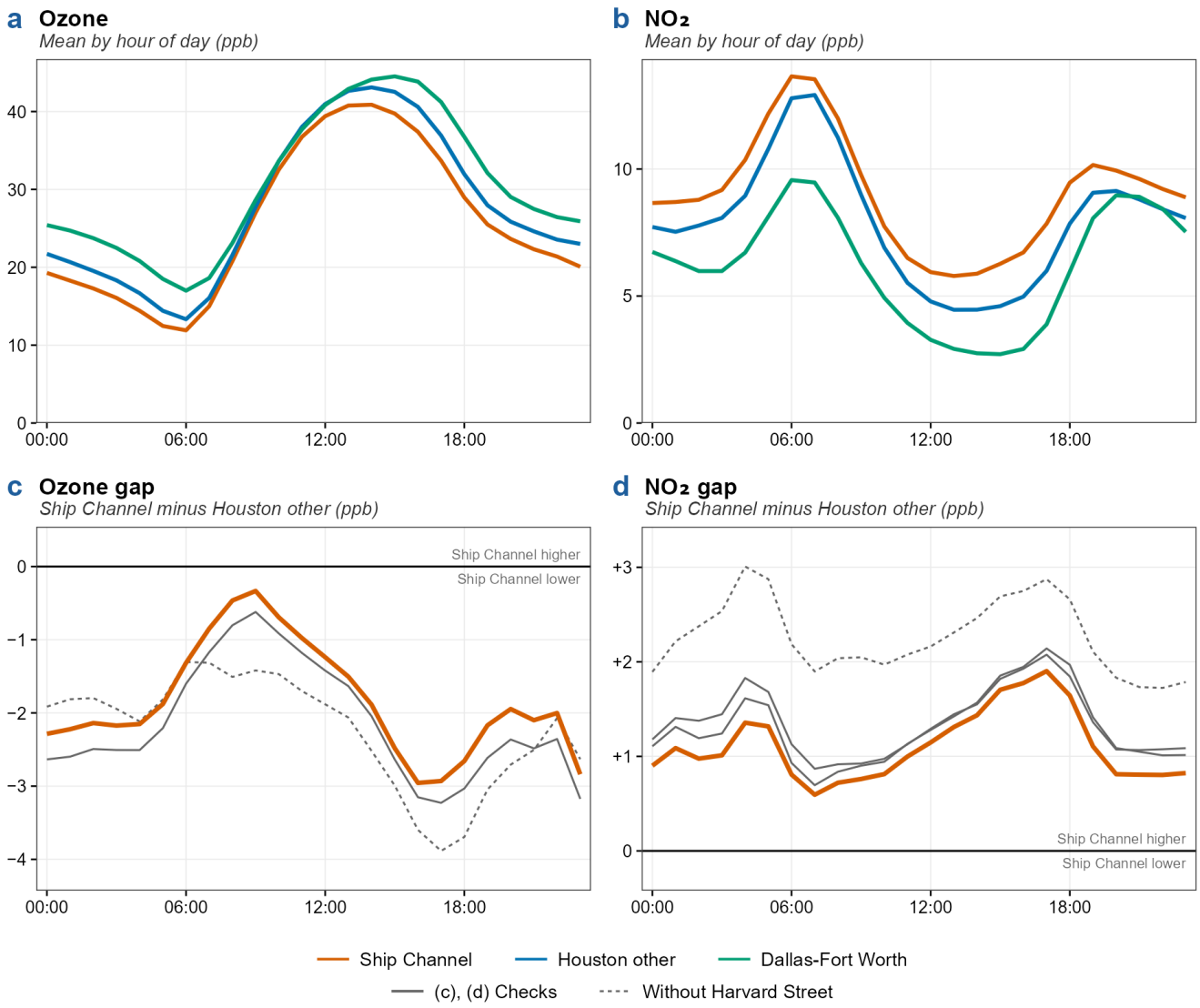
Source: EPA AQS daily data; five-year monitors (site and gas): Ship Channel 16, Houston other 15, Dallas-Fort Worth 13. Shaded: the events.csv window. Normal: the same dates in the other four years. SO_2 sits near zero and is left out of (b) and (d).

Figure 2B.6. Beryl knocked out most Ship Channel monitors.
Record: case_study_event_reporting.csv, case_study_event_levels.csv



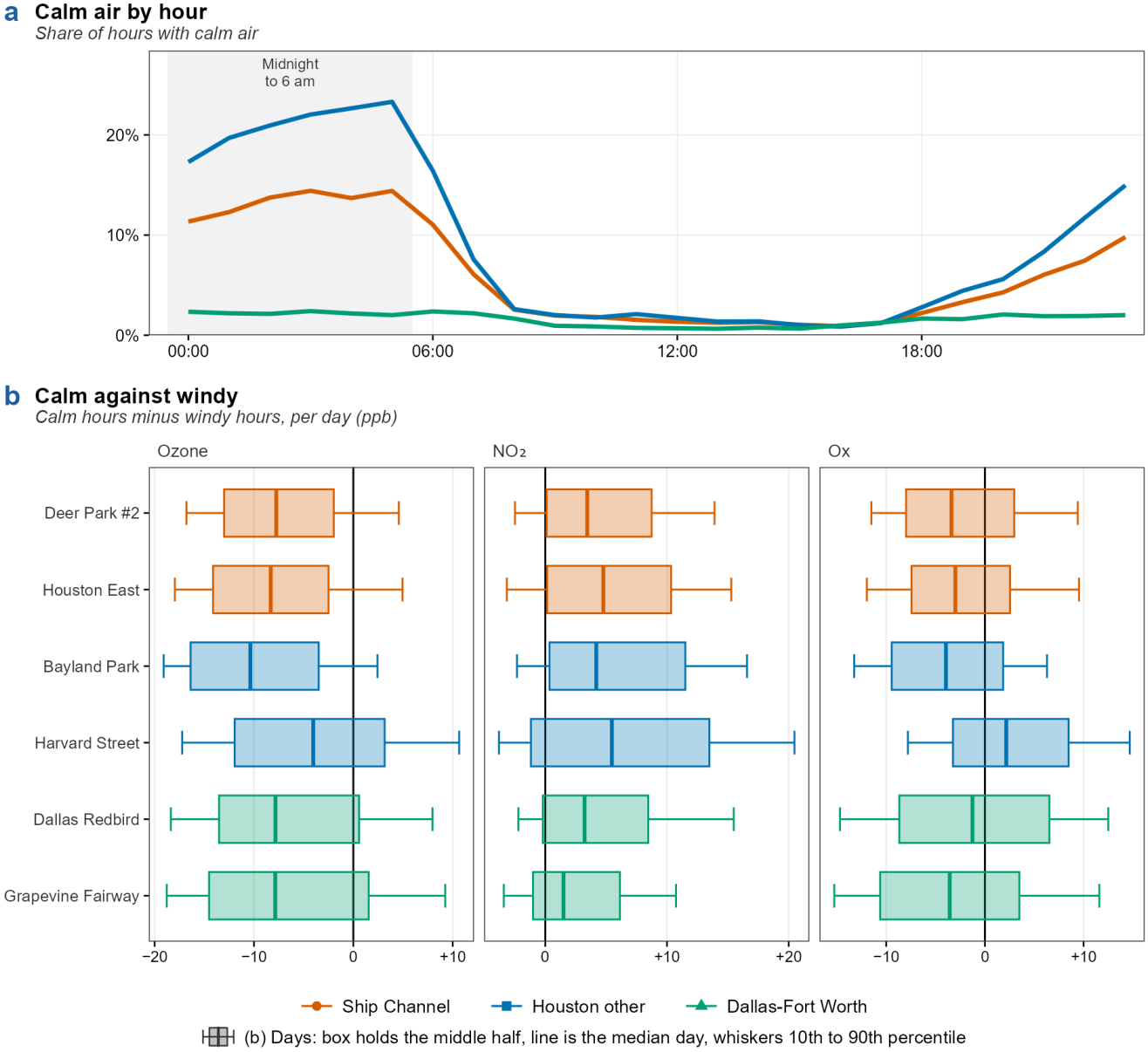
Source: EPA AQS daily data, 2021 to 2025, five-year sites. Group value: mean of the sites reporting that day.

Figure 2B.7. On the median day, NO_2 is higher at the Ship Channel.
Record: case_study_paired_differences.csv



Source: EPA AQS hourly data, 2021 to 2025, ordinary hours, 2 sites per group, each group the mean of its sites; local standard time (06:00 is 7 am daylight time). (c), (d) Every check keeps both gaps on the same side of zero at all hours.

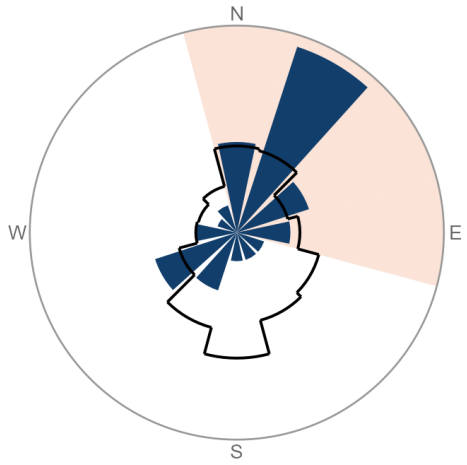
Figure 2C.2. Ship Channel NO_2 is higher and ozone lower at every hour.
Record: case_study_hourly_gaps.csv



Source: EPA AQS hourly data, 2021 to 2025, ordinary hours with wind; calm: under 0.5 m/s. (b) Against windy hours at the same site, hour and month; Ox = ozone + NO_2.

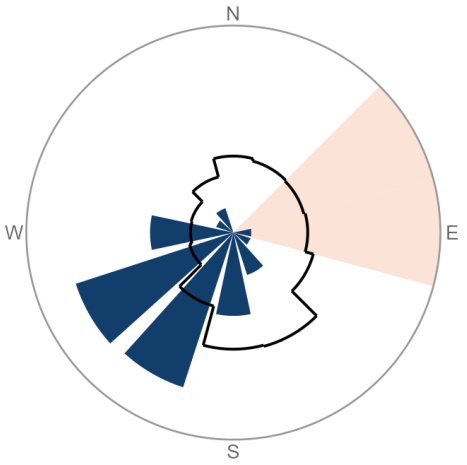
Figure 2C.8. Calm air lifts NO_2 and cuts ozone at every site.
Record: case_study_calm_air.csv

a **Park Place**
Ship Channel



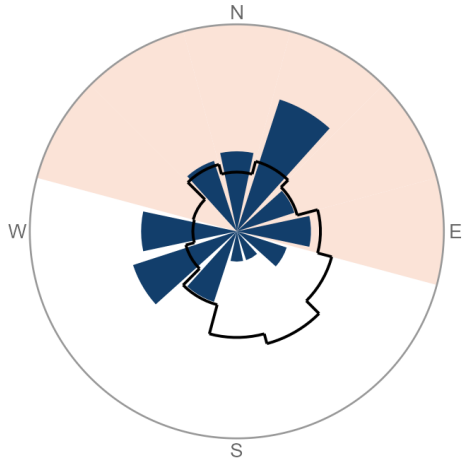
Channel side: 77% of spikes, 30% of hours

b **Croquet**
Houston other



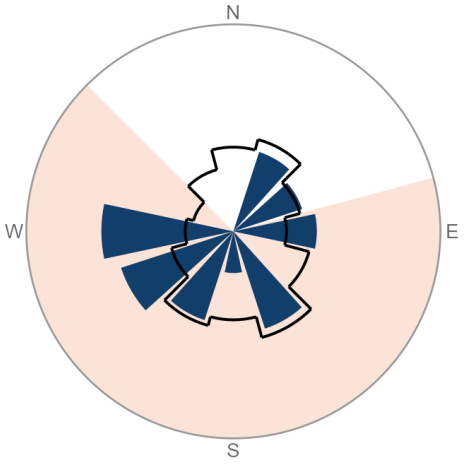
Southwest: 75% of spikes, 11% of hours

c **Deer Park #2**
Ship Channel



Channel side: 56% of spikes, 37% of hours

d **Clinton**
Ship Channel

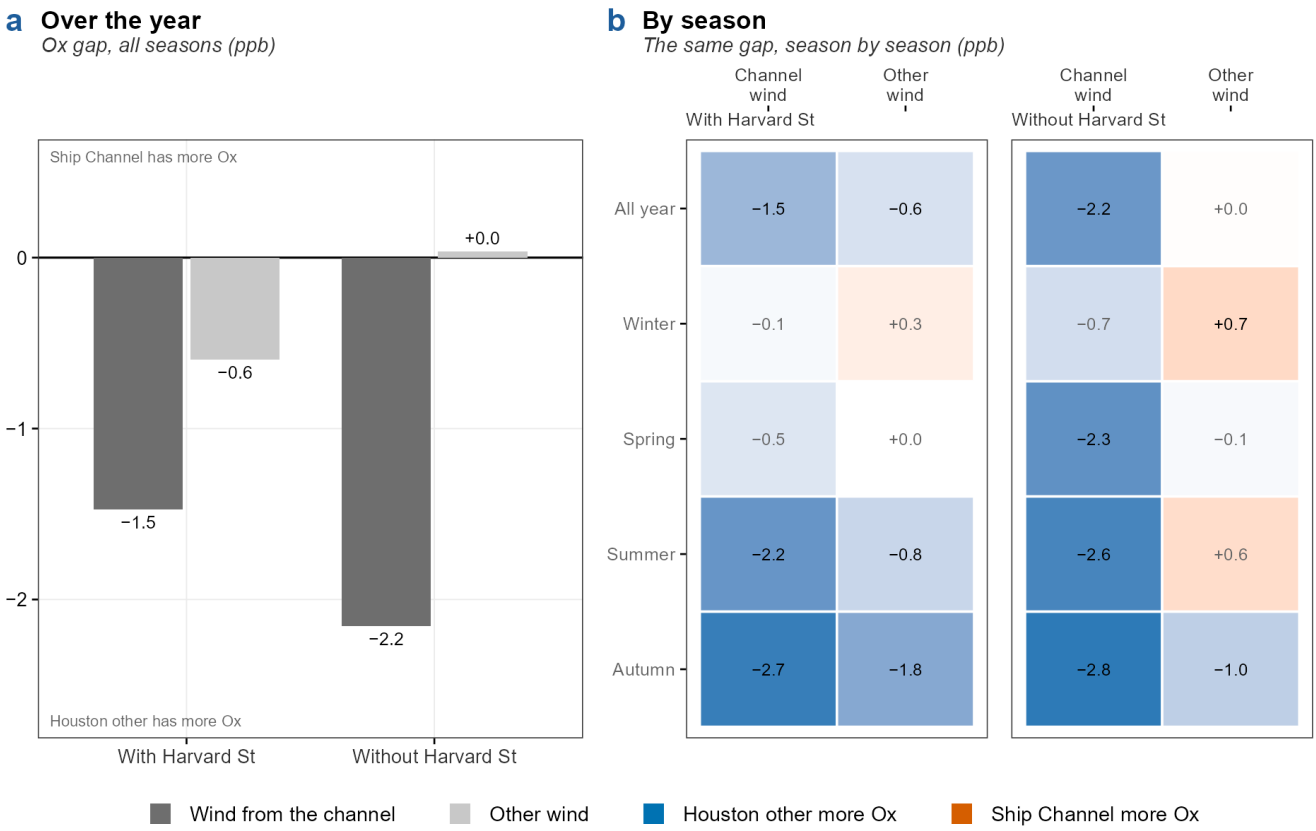


Channel side: 83% of spikes, 65% of hours

Spike hours — All hours — Toward the Ship Channel

Source: EPA AQS hourly data, 2021 to 2025, ordinary hours with a wind direction; 30° sectors. Spike: an hour at 5 ppb or more. Petal area shows each direction's share; past the outline, spikes come from that side more often than the wind does.

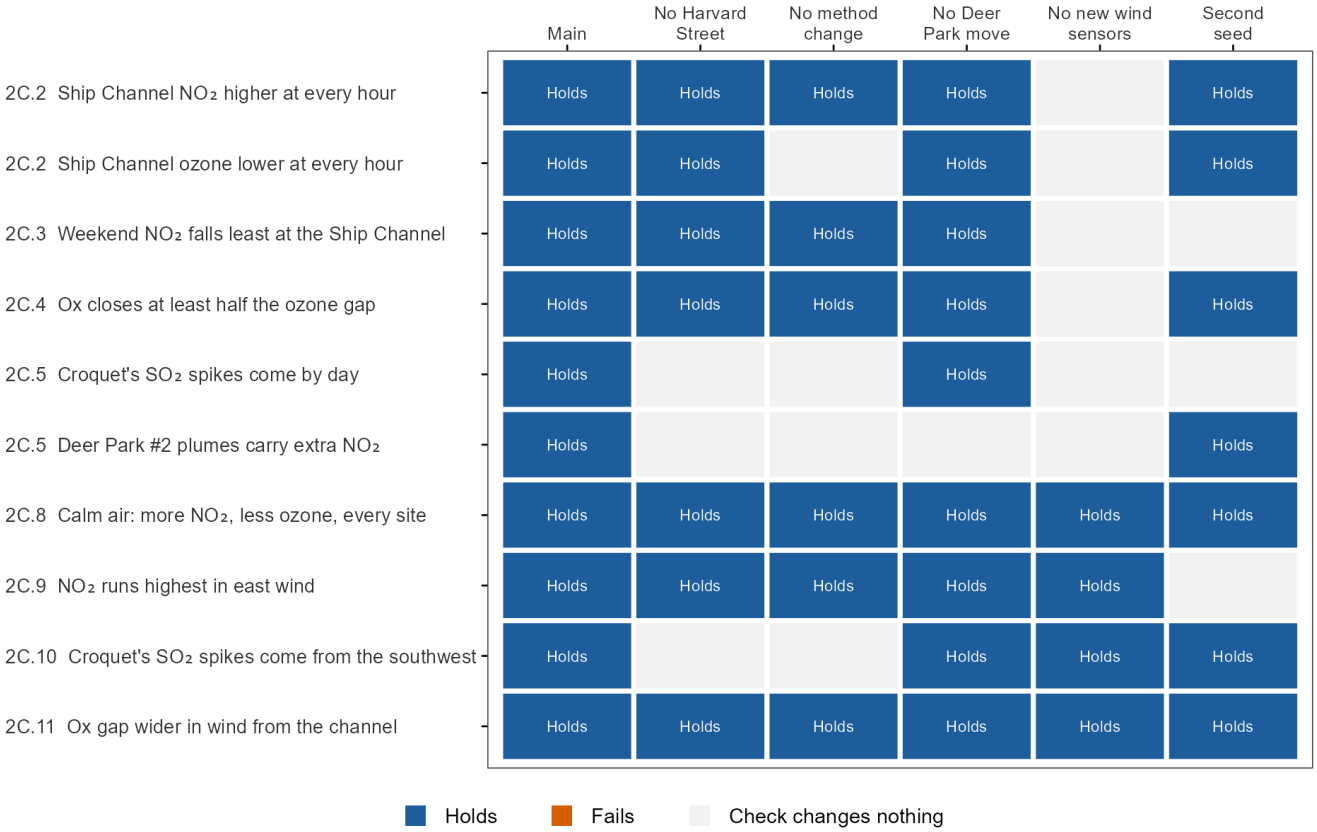
Figure 2C.10. Croquet's SO_2 spikes come from the southwest.
Record: case_study_channel_bearings.csv, case_study_so2_directions.csv



Source: EPA AQS hourly data, 2021 to 2025; wind: mean of the 6 Houston sensors, calm hours left out. Channel wind: 45° to 135°, where the channel lies from Bayland Park and Harvard Street. (b) Grey values: their 95% range includes zero.

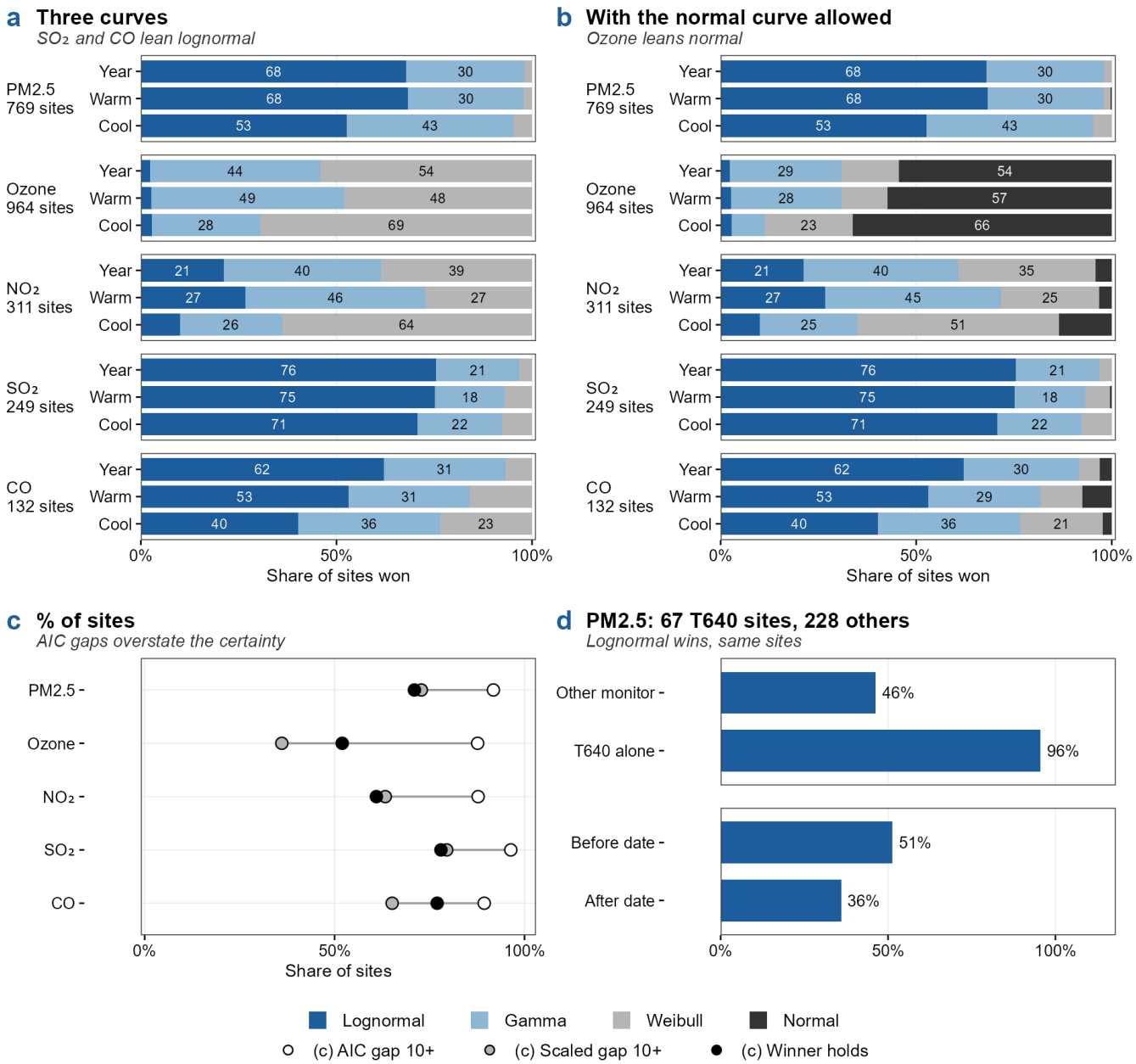
Figure 2C.11. The leftover Ox gap appears in wind from the channel.
Record: case_study_ox_gap_by_wind.csv

Findings and checks
Does each finding survive each check?



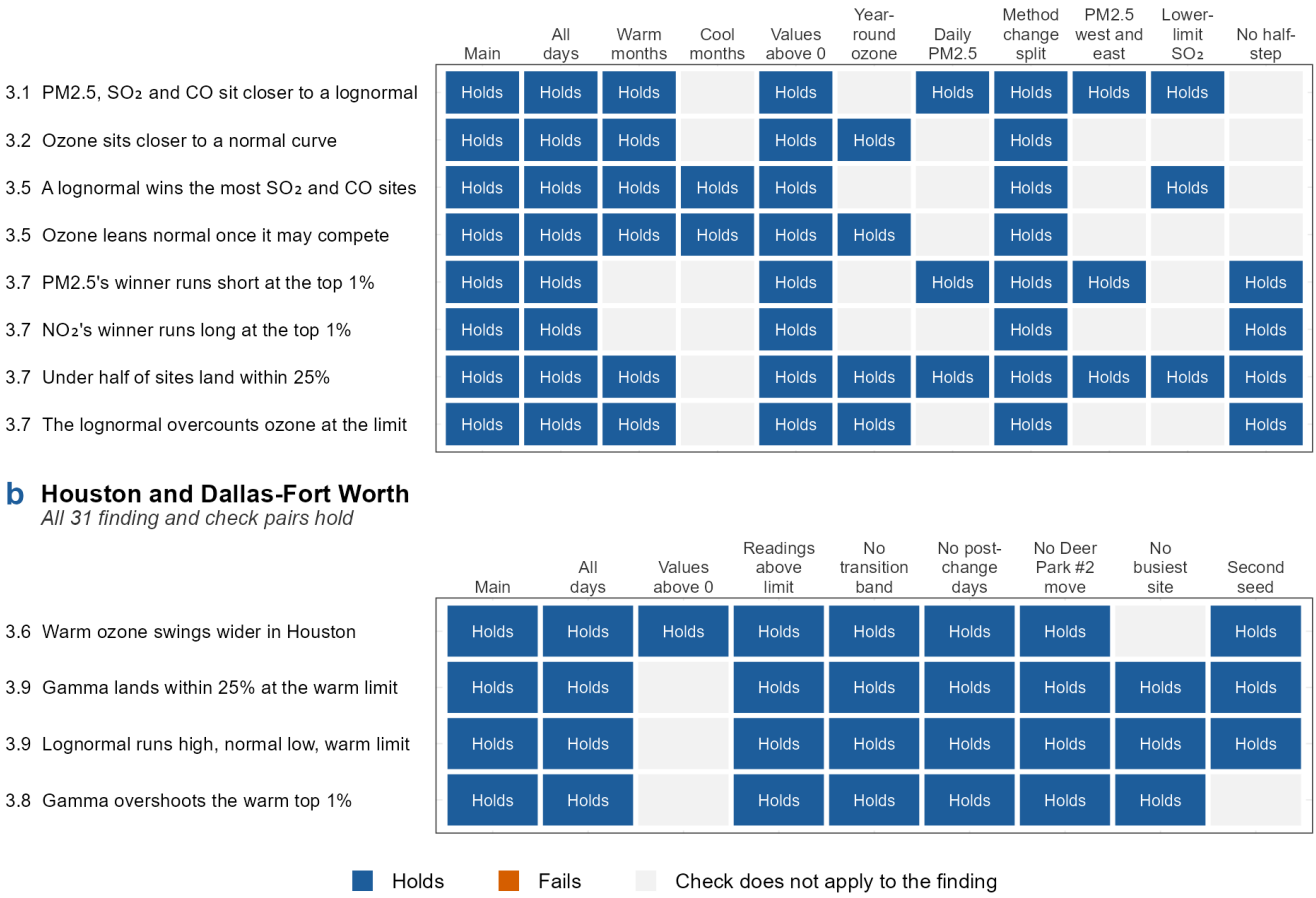
Checks: without Harvard Street, the nearest Houston other site; without its NO_2 after the April 2024 method change; without Deer Park #2 after its May 2025 move; without hours on new wind sensors. Second seed: 67 ranges redrawn with seed 2027.

Figure 2C.12. Every hourly finding holds under every check.
Record: case_study_hourly_robustness.csv, case_study_hourly_seed_check.csv



Source: EPA AQS daily data, 2021 to 2025, ordinary days, five-year sites. Winner: lowest AIC per site. (c) Holds: kept in 90%+ of 100 two-week block resamples of 100 random sites per pollutant. (d) Split at 2023-02-18 for sites without a T640.

Figure 3.5. SO_2 and CO lean lognormal and ozone leans normal.
Record: distribution_fit_summary.csv, distribution_fit_checks.csv



Source: Phase 3 records, EPA AQS daily data 2021 to 2025. (a) five-year sites outside the near-zero rule; (b) warm-month case-study ozone. Second seed: the 95% ranges redrawn with seed 2027.

Figure 3.10. Every distribution finding holds under every check.
Record: distribution_robustness.csv

A.3 Key code

These excerpts are read from the scripts each time the report is built, so they always match the code behind the results.

NAAQS screening (utils.R). Each site-year gets its standard's statistic (a ranked daily value, or the mean of four quarterly means for annual PM2.5), rounded the way the standard specifies, and is compared with the level.

```
naaqs_statistics <- function(site_days, naaqs, naaqs_ranks, percentile_ranks) {
  site_year_ranks <- site_days %>%
    count(pollutant, site_id, year, name = "n_days") %>%
    left_join(naaqs_ranks, by = "pollutant", relationship = "many-to-one") %>%
    left_join(percentile_ranks, by = join_by(percentile, between(n_days, min_days, max_days)),
        relationship = "many-to-one") %>%
    mutate(value_rank = coalesce(fixed_rank, rank)) %>%
    select(pollutant, site_id, year, n_days, value_rank)

  stopifnot(
```

```
    "Every site-year has a rank" = !anyNA(site_year_ranks$value_rank),
    "No rank is beyond the site-year's days" = all(site_year_ranks$value_rank <= site_year_ranks$n_days)
  )

  short_term <- site_days %>%
    arrange(pollutant, site_id, year, desc(value)) %>%
    mutate(value_order = row_number(), .by = c(pollutant, site_id, year)) %>%
    inner_join(site_year_ranks, by = c("pollutant", "site_id", "year"),
        relationship = "many-to-one") %>%
    filter(value_order == value_rank) %>%
    inner_join(filter(naaqs, standard != "Annual") %>% select(pollutant, standard),
        by = "pollutant", relationship = "many-to-one") %>%
    select(pollutant, standard, site_id, year, n_days, value)

  pm25_annual <- site_days %>%
    filter(pollutant == "PM2.5") %>%
    mutate(quarter = quarter(date_local)) %>%
    summarise(quarter_mean = mean(value), .by = c(pollutant, site_id, year, quarter)) %>%
    summarise(n_quarters = n(), value = mean(quarter_mean), .by = c(pollutant, site_id, year)) %>%
    mutate(standard = "Annual") %>%
    left_join(select(site_year_ranks, pollutant, site_id, year, n_days), by = c("pollutant",
        "site_id", "year"), relationship = "one-to-one")

  stopifnot(
    "One short-term value per site-year" = nrow(short_term) == nrow(site_year_ranks),
    "Annual PM2.5 has 1 to 4 quarters per site-year" = all(between(pm25_annual$n_quarters, 1, 4))
  )

  pm25_annual <- pm25_annual %>%
    filter(n_quarters == 4)

  bind_rows(short_term, pm25_annual) %>%
    left_join(select(naaqs, pollutant, standard, level, digits, rounding), by = c("pollutant",
        "standard"), relationship = "many-to-one") %>%
    mutate(value_rounded = if_else(rounding == "truncate", truncate_to(value, digits),
        round_half_up(value, digits))) %>%
    mutate(is_above = value_rounded > level) %>%
    select(pollutant, standard, site_id, year, n_days, n_quarters, value, value_rounded, level,
        is_above)
}
```

Chunked hourly read (utils.R). Each 1-million-row chunk is filtered to the case-study monitors before the next is read, so the full hourly files never sit in memory, and the result is cached until the zip changes.

```
keep_case_study_rows <- function(chunk, monitors) {
  chunk %>%
    filter(`State Code` == "48") %>%
    mutate(site_id = str_c(`State Code`, `County Code`, `Site Num`, sep = "-")) %>%
    semi_join(monitors, by = c("site_id", "Parameter Code" = "parameter_code"))
}

filter_case_study_zip <- function(zip_path, out_dir, monitors) {
  out_path <- file.path(out_dir, str_replace(basename(zip_path), "\\.zip$", ".rds"))

  zip_size <- file.size(zip_path)

  zip_mtime <- as.numeric(file.mtime(zip_path))

  if (file.exists(out_path)) {
    saved <- read_rds(out_path)
    if (identical(saved$zip_size, zip_size) && identical(saved$zip_mtime, zip_mtime)) {
```

```
      return(out_path)
    }
  }

  rows_read <- 0

  filtered_rows <- read_csv_chunked(
    zip_path,
    DataFrameCallback$new(\(chunk, pos) {
      stopifnot("readr reported parsing problems" = nrow(problems(chunk)) == 0)
      rows_read <<- rows_read + nrow(chunk)
      keep_case_study_rows(chunk, monitors)
    }),
    chunk_size = 1e6,
    col_types = aqs_hourly_col_spec
  )

  write_rds(
    list(rows_read = rows_read, data = filtered_rows, zip_size = zip_size, zip_mtime = zip_mtime),
    out_path,
    compress = "gz"
  )

  rm(filtered_rows)
  invisible(gc())

  out_path
}
```

Same-day differences and the block bootstrap (2b_daily_case_study.R). The median daily difference gets a 95% range from resampled 14-day blocks, which keeps the correlation between neighbouring days.

```
block_bootstrap_range <- function(x) {
  n <- length(x)
  if (n < 2 * bootstrap_block_days) {
    return(c(NA_real_, NA_real_))
  }
  block_starts <- seq_len(n - bootstrap_block_days + 1)
  n_blocks <- ceiling(n / bootstrap_block_days)
  medians <- vapply(seq_len(bootstrap_reps), \(rep) {
    picked <- sample(block_starts, n_blocks, replace = TRUE)
    median(x[as.vector(outer(seq_len(bootstrap_block_days) - 1, picked, `+`))][seq_len(n)])
  }, numeric(1))
  unname(quantile(medians, c(0.025, 0.975)))
}

summarise_differences <- function(differences, by) {
  differences %>%
    arrange(date_local) %>%
    summarise(
      n_days = n(),
      median_difference = median(difference),
      pct_days_positive = 100 * mean(difference > 0),
      range = list(block_bootstrap_range(difference)),
      .by = all_of(by)
    ) %>%
    mutate(ci_low = vapply(range, \(r) r[1], numeric(1))) %>%
    mutate(ci_high = vapply(range, \(r) r[2], numeric(1))) %>%
    select(-range)
}
```

Dropping checks that change nothing (2b_daily_case_study.R). A check whose days and values match the main run exactly would only add resampling noise, so it is removed before any range is drawn.

```
difference_series <- daily_differences %>%
  summarise(dates = list(date_local), values = list(difference), .by = c(site_set, contrast,
      pollutant))

main_series <- difference_series %>%
  filter(site_set == "Main") %>%
  select(contrast, pollutant, main_dates = dates, main_values = values)

unchanged_checks <- difference_series %>%
  filter(site_set != "Main") %>%
  inner_join(main_series, by = c("contrast", "pollutant"), relationship = "many-to-one") %>%
  mutate(is_unchanged = mapply(\(d, v, md, mv) identical(d, md) && isTRUE(all.equal(v, mv)), dates,
      values, main_dates, main_values)) %>%
  filter(is_unchanged) %>%
  select(site_set, contrast, pollutant)

print(unchanged_checks, n = Inf)

daily_differences <- daily_differences %>%
  anti_join(unchanged_checks, by = c("site_set", "contrast", "pollutant"))
```

The Ox balance (2c_hourly_eda.R). Ozone, NO_2 and Ox gaps are taken in the same hour, and the share of the ozone gap that Ox closes is one minus the Ox gap over the ozone gap.

```
ox_paired_gaps <- ox_long %>%
  filter(site_group != "Dallas-Fort Worth") %>%
  summarise(group_ppb = mean(value_ppb), .by = c(site_set, measure, site_group, datetime_utc,
      date_local, hour_local, season)) %>%
  mutate(site_group = if_else(site_group == "Houston Ship Channel", "ship_channel_ppb",
      "houston_other_ppb")) %>%
  pivot_wider(names_from = site_group, values_from = group_ppb) %>%
  filter(!is.na(ship_channel_ppb), !is.na(houston_other_ppb)) %>%
  mutate(gap_ppb = ship_channel_ppb - houston_other_ppb)

ox_checks <- ox_season_gaps %>%
  select(site_set, season, measure, n_days, gap_ppb, ci_low, ci_high) %>%
  mutate(measure = str_to_lower(measure)) %>%
  pivot_wider(names_from = measure, values_from = c(gap_ppb, ci_low, ci_high),
      names_glue = "{measure}_{.value}") %>%
  select(site_set, season, n_days, ozone_gap_ppb, no2_gap_ppb, ox_gap_ppb, ozone_ci_low,
      ozone_ci_high, ox_ci_low, ox_ci_high) %>%
  mutate(share_of_ozone_gap_closed = 1 - ox_gap_ppb / ozone_gap_ppb) %>%
  mutate(across(where(is.double), \(x) round(x, 3))) %>%
  arrange(season, site_set)
```